

Valid exposure markers

Low screening sensitivity; excess not readily explained by regional smoking gradients alone.

- Lung (esp. never-smoker subset)
- Brain / CNS
- Leukaemia & lymphoma
- Oesophageal
- Kidney (solvent-exposed cohorts)
- Bladder
- Pediatric cancers (no pack-years)
- Male breast (solvent-exposed military)

Pediatric and germline/sex-ratio signals are the most confounding-resistant of all.

Screening-sensitive / confounded

Incidence strongly shaped by detection, hormones, or behaviour — not primary markers.

- **Female breast — excluded as primary marker**
~37% apparent rise from screening lead-time alone; reproductive transition; uneven regional rollout. Retained only in radiation cohorts (dose-response).
- **Thyroid — interpret with caution**
Incidental detection inflates incidence; but is radiation-sensitive, so relevant in fallout cohorts.
- **Prostate — interpret with caution**
PSA-screening-driven; not tobacco-attributable, so a spatial cluster is still notable but ambiguous.
- **Colorectal — secondary outcome**
Common and registry-stable, but only modestly exposure-specific.

Excluding screening-sensitive sites from the primary marker set is a deliberate methodological choice that strengthens the analysis.